MKALAMA DISTRICT COUNCIL

FORM THREE TERMINAL EXAMNATION

CHEMISTRY

Time: 3 Hours **MAY 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C **thirty (30)** marks.
4. Non- programmable calculators may be used.
5. The following constants may be used
   H=1, C= 12, O= 16, Na=23, S=32, Cl= 35, K=39, Fe=55.8, Ca= 40, Mg =24, Ag = 108, Cu = 63.5,
   Avogadro's constant, (A)= 6.02×10^{23}
   GMV at S.T.P =22.4 $dm^3\ mol^{-1}$
   1 liter = $1dm^3$ =1000 cm^3

wazaelimu.com

SECTION A (16 Marks)

Answer **all** questions in this section.

1 For each of the items **(i) – (x),** choose the correct answer from the given alternatives and write its letter in the table provided below.

(i) During volumetric analysis in the school chemistry laboratory the solution added to the burette by a form three student is referred to as:

A. Acid and base
B. Secondary solution
C. Analyte
D. Titrant
E. Alkaline solution.

(ii) The valency of an element found at Geita mining industry with atomic number 10 is :-

A. 2 B. 1 C. 3 D. 0 E. 4

(iii) Which method could be used to separate the products in the following chemical reaction?

$Pb(NO)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)} + 2KNO_{3(aq)}$

A. condensation B. filtration C. Chromatography D. distillation E. separation funnel

(iv) 10 cm^3 of 0.4M sodium hydroxide are added to 40cm^3 of 0.2M hydrochloric acid. The resulting mixture will be:-

A. Alkaline B. Neutral C. acidic D. Strong alkaline E. Weak acidic

(v) Mr . Ndemanga mixed two solutions in a boiling tube at 20°C. After mixing the two solutions the new products were found to have a temperature of 10°C. This shows that the reaction was;

A. An exothermic B. An endothermic C. Enthalpy D. neutralization E. combustion

(vi) What is the IUPAC name of H_2SO_3

A. Sulphuric (iv) acid
B. Sulphuric (vi) acid
C. Sulphuric acid
D. Hydrogen sulphate
E. Hydrogen trisulphate

(vii) Which of the following sets of process use a gas that ignites with a pop sound when a lighted splint is passed through it?

A. Ballon filling, welding and dividing
B. Hardening of oil, ballon filling and diving
C. Hardening of oil, ballon filling and diving
D. Fueling of rocket, diving and welding
E. Fueling car, diving and welding.

(viii) Mr. Machepe Abdallah was demonstrating an experiment by dissolving sugar in water until the solute was not dissolving any more, the resulting mixture is;

A. Emulsion B. Saturated solution C. Suspension D. Unsaturated Solution E. Super Saturated solution

(ix) Image your brother was preparing dinner in the kitchen, he started chopping potatoes into small pieces to make chips, what was the purpose of this action?

A. Make them delicious
B. Make them dry
C. Shorten the cooking time
D. To increase the amount of potato chips
E. To increase concentration of potatoes.

(x) Which of the following substances where its dissolution in water causes permanent hardness of water?

A. Calcium sulphate
B. Magnesium carbonate
C. Sodium Chloride
D. Calcium Carbonate
E. Sodium Carbonate

2 Match the type of reaction in **List A** with a correct response in **List B** by writing the letter of the correct response below the corresponding item number in answer sheet provided .

LIST A	**LIST B**
(i) Decomposition reaction	A: Shows only chemical species that involved in the reaction.
(ii) Displacement reaction	B: $F_2 + 2HCl \rightarrow 2HF + Cl_2$
(iii) Net ionic equation	C: Complex chemical compound broken into simple molecules.
(iv) Neutralization reaction.	D: Combination reaction.
(v) Complete ionic equation	E: Double decomposition reaction.
(vi) $2H_2(g) + O_2(g) \rightarrow 2H_2O$	F: Both spectator ions and reacting species are included.
	G: Found both sides of the chemical equation.
	H: Molecular formula
	I: Only hydroxide ions and hydrogen ions are involved

SECTION B (54 Marks)

Answer **all**

3 (a) Hydrogen is most useful element in the world, you as chemist explain how hydrogen is made in the laboratory and industry (use chemical equations)

(b) Explain how does hydrogen react with copper (II) oxide? (Chemical equation is necessary)

(c) Hydrogen is a promising fuel for the future. Justify this statement by giving two (2) reasons.

(d) Most uses of hydrogen are dictated by its properties. Explain this statement by giving four (4) reasons

4 (a) The hydroxide of metal M react with HCl as follows

$M(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow MCl_{2(aq)} + 2H_2O_{(l)}$

It was found out that 5.8g of $M(OH)_2$ react with 0.2 M HCl. Calculate the

i. Atomic mass of metal M.
ii. Suggest the name of metal M.

5 Amina, Mwanaidi, Godfrey, Ntimizi and Magreth and Ester Performed an experiment to determine hardness of water by using soap and the following water sample.
Water sample A=water from lembeli
B=water from mtakuja
C=water from kiloloeli
G=water from ujega

Sample	VOLUME OF SOAP SOLUTION USED (cm^3)		
	Untreated water	**boiled water**	**Water passed in the ion exchanger**
A	12	1.8	1.8
B	17	17	1.7
C	26	20	1.8
D	1.6	1.6	1.6

(a) Which water sample is hardest? Give reason for your answer.
(b) Which water sample behave as distilled water? Explain.
(c) Name a chemical substance that could be that cause of hardness in
(i) water sample A
(ii) water sample
(d) which problems may face people who are using water from lembeli, mtakuja and kjiloleli?

6 (a) Form three students visited chemistry laboratory and laboratory technician insisted them to obey all statements that explain what to do or not to do before, during and after practical activities. Give three (3) reasons why should these statements be obeyed by students in the laboratory?
(b) Give one reason to support each of the following statements
i. A laboratory should be equipped with working fire extinguisher's
ii. A laboratory should have large windows iii) A laboratory should have fume chamber
(c) i. In which other areas do you find the safety signs? (two (2) points)
ii. Briefly explain why are the safety signs so important in the laboratory?

7 (a) What volume of hydrogen gas is produced at STP when 0.65g of zinc reacts completely with excess hydrochloric acid?
(b) What mass of sulphuric acid is found in 400 cm^3 of its 0.1M aqueous solution?
(c) What is the molarity of :-
(i)solution containing 5.3g of sodium carbonate in 100ml of the solution?
(ii) solution containing 80g of sodium hydroxide in 2 litre?

8 (a) One of the form one student decided to separate the mixture of water and sand which are mixed accidently, use this information to respond on the questions below;
i. What are the apparati/equipment's which can be used to separate the above mixture (Give three points).
ii. By using a well labeled diagram explain how those two components can be separated by this student.
(b) Briefly explain how are solutions, suspensions and emulsions important in our life? (Give three points)
(c) Form three students observed that water is changing to vapour after being boiled to 100ºC. You as chemist, explain how the process of changing water to vapour is used at home. (Give three (3)

SECTION C (30 Marks)

Answer **two (2)** questions in the section.

9 A doctor from Muhimbili Hospital prescribed antacids tablets made from magnesium hydroxide to a student who was suffering from excess stomach acid

(a) Why did the doctor prescribe the antacid tablets?

(b) Write a balanced chemical equation that took place in the student's stomach after taking the tablets.

(c) If the normal acid content in the stomach is 160millimolar and the stomach secrets 0.5litres of gastric juice in a day, calculate the mass of the tablets to be given to the student if the gastric juice had 210millimolar hydrochloric acid.

(d) If each tablet of an antacid weighs 145mg of magnesium hydroxide, how many tablets will the students be given?

(e) If a dose requires 1 tablet per every 12hours, how many days should the student take to finish the full dose?

1 (a) One of form three students was assigned a task of cutting the branches of trees around school laboratory. Accidentally a bee stung a student on his arm. Perhaps a teacher assisted him by applying Ash powder on it.

i. mention the type of reaction which was expected to take place when ashes were applied

ii. Give out other four (4) application of the reaction occurred in a (i) above in our daily life

(b) With the aid of a balanced chemical equation, suggest the best methods for preparing the following salts in the laboratory.

i. Lead (II) sulphate

ii. Copper (II) nitrate

iii. Calcium Chloride

iv. Calcium Carbonate

v. Barium Sulphate

1 Most of farmers in the Matombo Village were interested to know how the knowledge of chemistry is necessary in agricultural activities. Assume you are chemist; educate these farmers on the application of chemistry in agriculture by giving five points